Ahmed Iprahim Abdelhamed

Data Scientist | Machine Learning, Deep Learning & API Deployment

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SUMMARY

Data Scientist with hands-on experience building end-to-end machine learning pipelines, deep learning models, and interactive deployment interfaces. Skilled in Python, Scikit-Learn, TensorFlow/Keras, PyTorch, XGBoost, SQL, and FastAPI foundations, with a focus on leakage-safe preprocessing, rigorous evaluation, and modular code structure. Currently advancing computer vision and API-based model serving capabilities through structured labs and portfolio projects.

CORE TECHNICAL SKILLS

Machine Learning & Deep Learning: XGBoost, Scikit-Learn, SVD, TensorFlow/Keras, PyTorch, Functional API, residual blocks, tensors, autograd, hyperparameter optimization, CNN/transfer-learning foundations in progress.

Feature Engineering & Diagnostics: ColumnTransformer, smoothed target encoding, cascade imputation, missingness analysis, MAR/MCAR diagnostics, multicollinearity analysis, VIF, data leakage prevention, train/validation/test separation.

Data Engineering & SQL: ETL pipelines, Kimball star schema, PostgreSQL, DuckDB, SQL querying, data validation, JSON parsing, public API data ingestion.

APIs & Backend: FastAPI, REST API fundamentals, path parameters, query parameters, Pydantic models, request/response validation, dependency injection, async endpoints, `requests`, API-based data collection.

Visualization & Deployment: Streamlit, Power BI, DAX, Looker Studio, Plotly, Matplotlib, Seaborn.

Evaluation Metrics: Precision-Recall AUC, ROC-AUC, RMSE, R^2, cross-validation, classification reports, confusion matrix analysis.

PROFESSIONAL EXPERIENCE

Tabby — Data & Analytics Specialist (Fintech Operations) 01/2024 – Present | Cairo, Egypt

- Engineered Python data pipelines to extract, clean, and validate multi-source event logs containing 50,000+ daily records, reducing manual processing overhead by approximately 80%.
- Established feature pipelines and anomaly checks to improve data reliability for operational and risk-related analysis.
- Queried relational databases to identify recurring delinquency patterns and escalated behavioral anomalies to Credit Risk teams.
- Investigated unauthorized disputes and Account Takeover cases using internal administrative tools and payment operations workflows.
- Automated recurring reporting workflows using Python, SQL, and BI tools, reducing manual reporting effort and improving turnaround time.

Independent Practice — Data Scientist & Risk Analyst (Freelance) 2023 – Present | Remote

- Delivered end-to-end data pipelines and predictive analytics solutions for financial lending datasets.
- Architected Kimball star schema data models and automated ETL workflows to ingest, clean, and consolidate disparate transaction datasets for downstream machine learning applications.
- Collected and normalized external data from public REST APIs using Python `requests` and JSON parsing to support feature engineering and model input preparation.
- Built reproducible ML workflows with emphasis on leakage prevention, validation integrity, and deployable model interfaces.

MACHINE LEARNING & DEEP LEARNING PROJECTS

Large-Scale Vehicle Valuation Engine — Deep Tabular ResNet

Tech Stack: TensorFlow/Keras, Python, Scikit-Learn, Missingno

- Architected a custom Deep Tabular ResNet using the Keras Functional API with residual skip connections and bottleneck projections, achieving an R^2 of 0.777 on 360,000+ real-world vehicle listings.
- Designed a 3-tier cascade imputation engine leveraging Missing at Random analysis and domain heuristics to handle nested missingness.
- Built a multi-stage preprocessing pipeline integrating smoothed target encoding, one-hot encoding, and RobustScaler to reduce target skewness and prevent train/validation/test data leakage.

Credit Risk & Expected Loss AI Engine

Tech Stack: Python, XGBoost, Scikit-Learn, PostgreSQL, Streamlit

- Engineered a credit classification pipeline evaluating Probability of Default and Expected Loss across 32,000+ borrower records while strictly excluding post-outcome features to prevent data leakage.
- Addressed severe class imbalance of 78:22 by optimizing XGBoost scale_pos_weight and tuning hyperparameters, increasing default recall to 78% without materially compromising precision.
- Deployed the predictive model as an interactive Streamlit web application enabling instant applicant scoring, automated risk tier assignment, and real-time simulations for underwriters.

CineMatch AI — Hybrid Recommender System

Tech Stack: Python, Scikit-Learn, NumPy, Pandas, Streamlit

- Developed a hybrid recommendation engine processing 10,000,000+ MovieLens ratings using Singular Value Decomposition with user/item bias and latent factor modeling.
- Achieved an RMSE of 0.8794, outperforming baseline models by 17% through feature engineering and hyperparameter tuning.
- Deployed a responsive Streamlit application featuring fuzzy string matching and real-time cosine similarity search.

Ames Housing Valuation & Regression Diagnostics

Tech Stack: Python, Scikit-Learn, Statsmodels, Streamlit

- Built an end-to-end regression pipeline predicting real estate valuations while optimizing model complexity and signal density.
- Reduced the feature space from 37 to 24 variables by diagnosing correlation matrices and Variance Inflation Factors to eliminate 13 multicollinear features.
- Validated LINE regression assumptions and engineered interaction/polynomial features, improving R^2 from 0.8356 to 0.8999.
- Deployed an interactive Streamlit application enabling real-time valuation estimates for stakeholders.

EDUCATION & CERTIFICATIONS

Bachelor of Science in Computer Science — Helwan University | Graduated 2024

Certifications: Advanced AI & Deep Learning Specialization (In Progress, 2026) | ALX Data Science Certificate (2024) | ALX Machine Learning Certificate (2024) | ALX Data Analytics Professional Certificate (2024) | ALX Python Programming Certificate (2024)